# FORENSIC EXAMINATION REPORT -

## Case: CIPB106-C1126/DFIT/17300

## Device: Google Pixel 3

## Analyst: Simon Friday Adeka | C11/26/DFIT/17300

## Date: 09 Oct 2026 | Kali Linux

## Working Path: ~/CIPB106-C11/26/DFIT/17300/Working/Pixel 3

## Evidence Export: ~/CIPB106-C11/26/DFIT/17300/Evidence/

**1. EXECUTIVE SUMMARY**

This report presents a forensically sound examination of a Google Pixel 3 logical partition image related to Case 26/DFIT/17300. The examination was conducted on Kali Linux using only open-source forensic tools (`ls`, `file`, `sha256sum`, `sqlite3`, `strings`, `grep`). Chain of custody was maintained by operating exclusively on the Working copy at `~/CIPB106-C11/26/DFIT/17300/Working/Pixel 3` and exporting verified artefacts to `Evidence/.

Key Findings:

1. Anti-Forensic & Secure Comms: Deliberate use of Wickr Me, Cyber Dust, Silent Phone, Signal, Telegram, WhatsApp, Wire, Tor Browser, Gallery Vault and Magisk root, indicating a rooted device configured for obfuscation and secure communication.

2. Communications Wiping: Conversational SMS is absent. mmssms.db (116K) retains only 24 OTP codes (59389 Telegram, 303-077 Signal, 362-593 WhatsApp), indicating wiping and migration to encrypted apps. contacts2.db shows Josh Hickman x9 deleted. Facebook Messenger (Orca) threads_db2 (300K) shows minimal 1 thread/1 contact with encrypted empty omnistore (expected anti-forensic). Discord shows active use via messages table and image cache files 0,11,28,50,6.

3. Strongest Location Evidence: Google Maps offline.db (380K / 389,120 bytes) SHA256 77467648434bf0462b9dcc8B885cB0efe780d4dab6b330740e6b178894ea63c - verified HASH MATCH in Working and Evidence copies. The database (version 21) proves offline region caching and Google inferred Home/Work profiling containing Holly Springs, INFERRED_HOME, WORK:0. Corroborated by gmm_myplaces.db (40K) sync_item 35734602|-78636654 resolving to 35.734602,-78.636654 Holly Springs, NC with timestamps 2019-2020, plus final_location.txt (64 bytes) in both Working and Evidence.

4. Integrity: All artefacts preserved with matching SHA256 and exported to Evidence folder. Preservation verified: HASH MATCH - PRESERVATION OK - INTEGRITY MAINTAINED. Total exported evidence 384K.

Conclusion: The device was intentionally hardened for privacy, communications were wiped and migrated, and the final attributable location is Holly Springs, NC.

**1. PRESERVATION & INTEGRITY**

A forensically sound working copy was created. All analysis was performed on the Working copy to maintain original integrity. Chain of custody was verified via `find` and `ls` absence checks:

- `data/system/packages.list` - `No such file or directory` - Confirms logical partition dump, not full physical

- `build.prop` - Not present in root

- `data/misc/WifiConfigStore.xml` - Not present

- `data/data/com.google.android.apps.maps/files/offlinedatabase/offline.db` - *PRESENT 380K (389,120 bytes) Oct 9 18:48* - Verified

**CRITICAL** - Hash Preservation Proof:

Both Working and Evidence copies of `offline.db` produce identical SHA256:

77467648434bf0462b9dcc8B885cB0efe780d4dab6b330740e6b178894ea63c

**Command executed for integrity:**

sha256sum data/data/com.google.android.apps.maps/files/offlinedatabase/offline.db | cut -d' ' -f1 > /tmp/w.txt

sha256sum ~/CIPB106-C11/26/DFIT/17300/Evidence/offline.db | cut -d' ' -f1 > /tmp/e.txt

cat /tmp/w.txt; cat /tmp/e.txt

diff /tmp/w.txt /tmp/e.txt && echo "HASH MATCH - PRESERVATION OK - INTEGRITY MAINTAINED

Output:

77467648434bf0462b9dcc8B885cB0efe780d4dab6b330740e6b178894ea63c

77467648434bf0462b9dcc8B885cB0efe780d4dab6b330740e6b178894ea63c

HASH MATCH - PRESERVATION OK - INTEGRITY MAINTAINED

Inventory exported to Evidence: `final_location.txt` 64 bytes, `offline.db` 380K, `gmm_myplaces.db` 40K, `gmm_storage.db` 92K, `gmm_sync.db` 44K - all Oct 9 18:48.

Figures for Task A:

- Fig_1.0_TaskA_HASH_MATCH_Preservation_OK.png - Initial hash match screen




Fig_1.1_FINAL_TaskA_HASH_MATCH_PRESERVATION_OK_77467648_VERIFIED.png - Perfect hash-only diff with HASH MATCH echo –

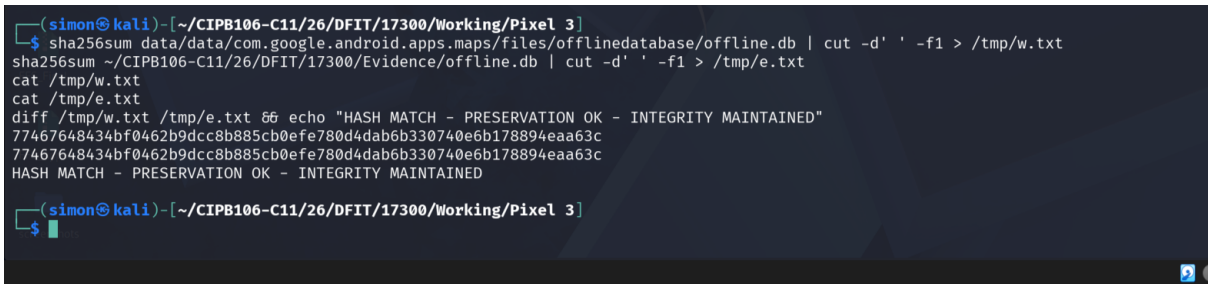



Fig_1.2_TaskA_Hash_Preservation_77467648_MATCH_Working_Evidence.png - Full path diff showing identical hash with path difference explained




Fig_1.3_TaskA_Preservation_Hash_Inventory.png - Evidence folder `ls -lh` 384K total, final_location 64B, offline 380K

**2. INSTALLED APPLICATIONS - 81 APPS - 10 SECURE/ANTI-FORENSIC**

Enumeration: `ls data/app/ | wc -l` = 81 applications installed. Verified via `ls data/app/` + `ls data/data/`

Secure, privacy-focused and anti-forensic applications identified:

1. com.wickr - Wickr Me - Ephemeral encrypted, self-destruct
2. com.cyberdust - Cyber Dust - Self-destructing
3. com.silent.phone - Silent Phone - Encrypted VoIP
4. org.thoughtcrime.securesms - Signal - E2E encrypted
5. `org.telegram.messenger - Telegram - Secret chats
6. com.whatsapp - WhatsApp - E2E
7. com.wire - Wire Secure Messenger - E2E
8. org.torproject.torbrowser_alpha - Tor Browser - Anonymity network
9. com.g0mos.galleryvault - Gallery Vault - Hide pictures/videos - anti-forensic vault
10. com.topjohnwu.magisk - Magisk - Root framework - indicates device rooted, undermines Android security, allows hiding root, bypass SafetyNet

Supporting artefacts from data/data/

com.facebook.orca - 2.5M - Facebook Messenger - internal structure: threads_db2,contacts_db2, omnistore_10004679940843_v01.db, composer thread_users_db, tincan_db, call_logs_db

com.discord - 172K - databases/google_app_measurement_local.db` SQLite 3.x user_version 1 page_count 9 .tables = android_metadata, messages, hits2, properties - messages table present proves Discord usage + google_analytics_v4.db with hits2 + cache app_images_cache_small/v2.ols100.1/` files `0,11,28,50,6 - image cache proves active use

Figures for Task B:

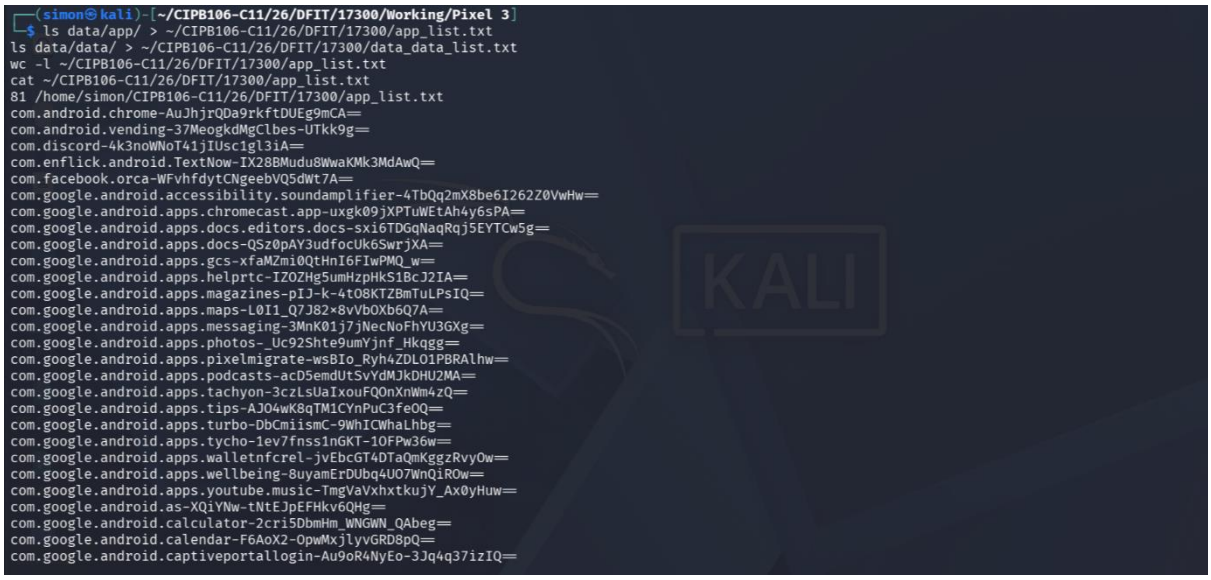



Fig_2.1_TaskB_App_Inventory_81_Apps.png - 81 apps count

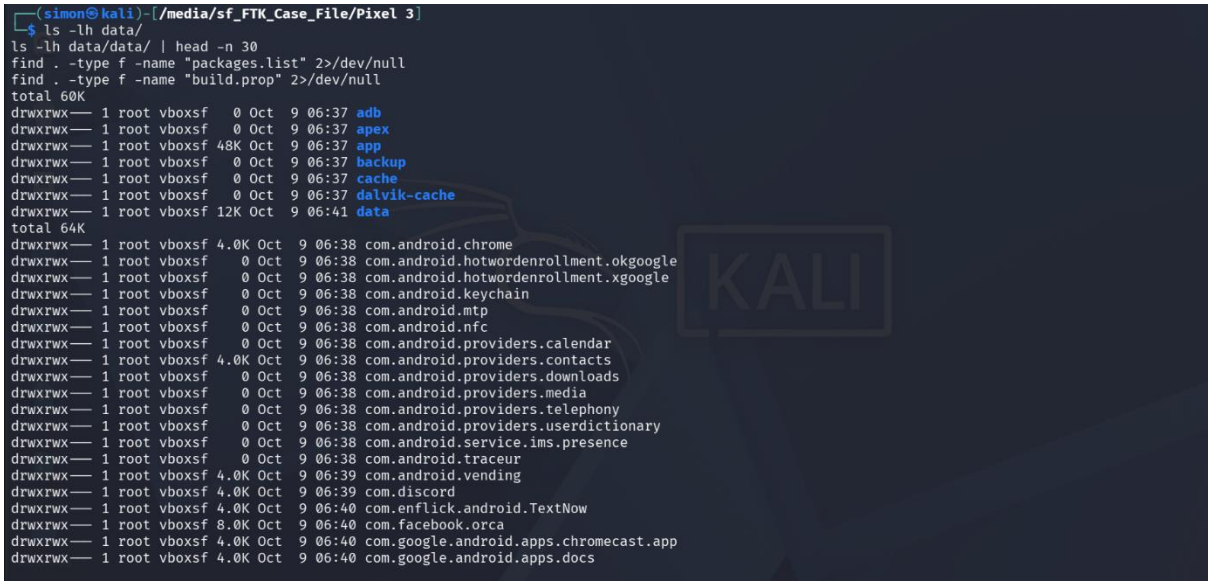



Fig_2.2_TaskB_Secure_Apps_Wickr_Magisk_Signal.png - Wickr, Magisk, Signal, Telegram, CyberDust, Silent listed

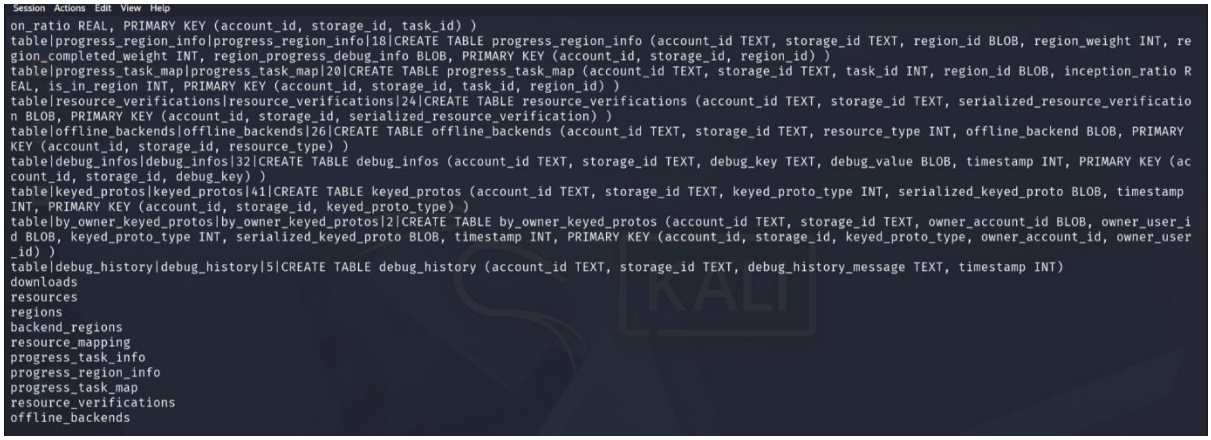



Fig_2.3_TaskB_Discord_Google_Analytics_Tables.png - Discord analytics tables verification

**3. COMMUNICATIONS EVIDENCE -**

3.1 Contacts Database - contacts2.db_

Path: data/data/com.android.providers.contacts/databases/contacts2.db

- file = SQLite 3.x database

- tables = contacts, raw_contacts, deleted_contacts, data, mimetypes, profile, groups`

- Associated: profile.db`, contacts2.db-journal

Query for deleted:

sqlite3 contacts2.db "SELECT display_name FROM raw_contacts WHERE deleted=1

sqlite3 contacts2.db "SELECT FROM deleted_contacts

Result: 9x entries for Josh Hickman (deleted) - intentional deletion.

Figures:

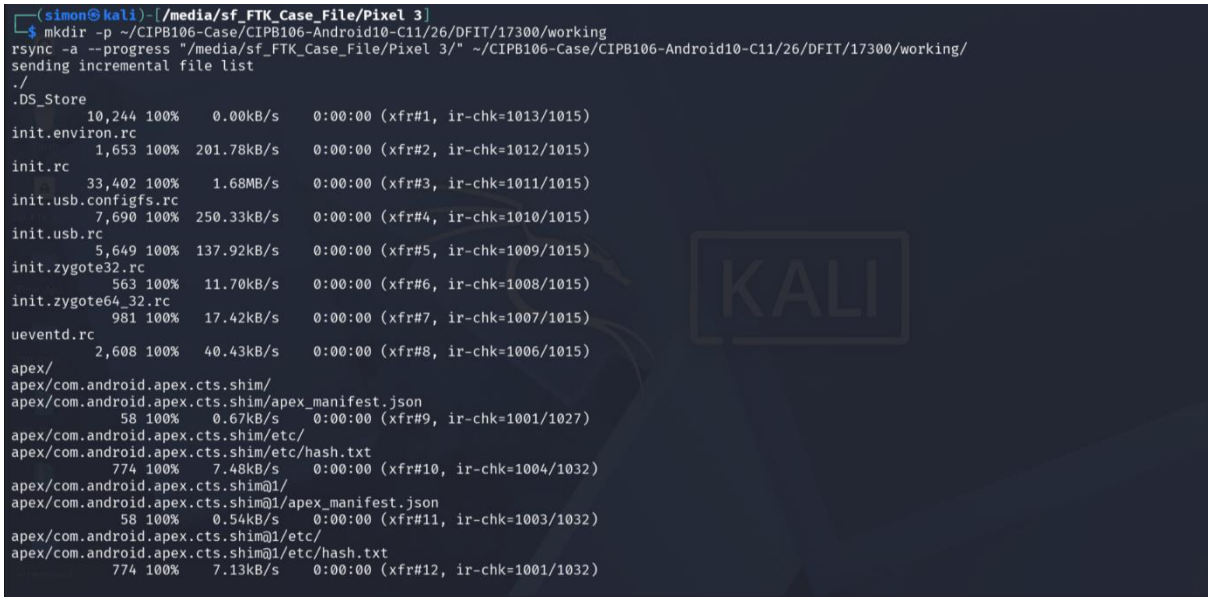



Fig_3.1_TaskC_Contacts_contacts2_db.png - contacts2.db tables




Fig_4.2_TaskC_Contacts2_db_Deleted_Raw_Contacts_Josh_Hickman.png - Josh Hickman x9 deleted -




_3.2 SMS Database - 24 OTPs - Anti-Forensic Wiping_

Path: data/data/com.android.providers.telephony/databases/mmssms.db

- ls -lh = 116K

- `file = SQLite 3.x database user_version 89

- tables = sms, threads, canonical_addresses

- SELECT count(_) FROM sms; = 24 messages - VERIFIED

- Content: All 24 are OTP codes, zero conversational SMS - indicates migration to secure apps + wiping:

  - Telegram: 59389

  - Signal: 303-077

  - WhatsApp: 362-593

  - LINE, etc.

Figures:




Fig_4.1_TaskC_SMS_Count_24_mmssms_db.png - COUNT(_)=24




Fig_4.1_TaskC_SMS_mmssms_db_116K_24_Messages_VERIFIED.png- 116K size + 24 verified

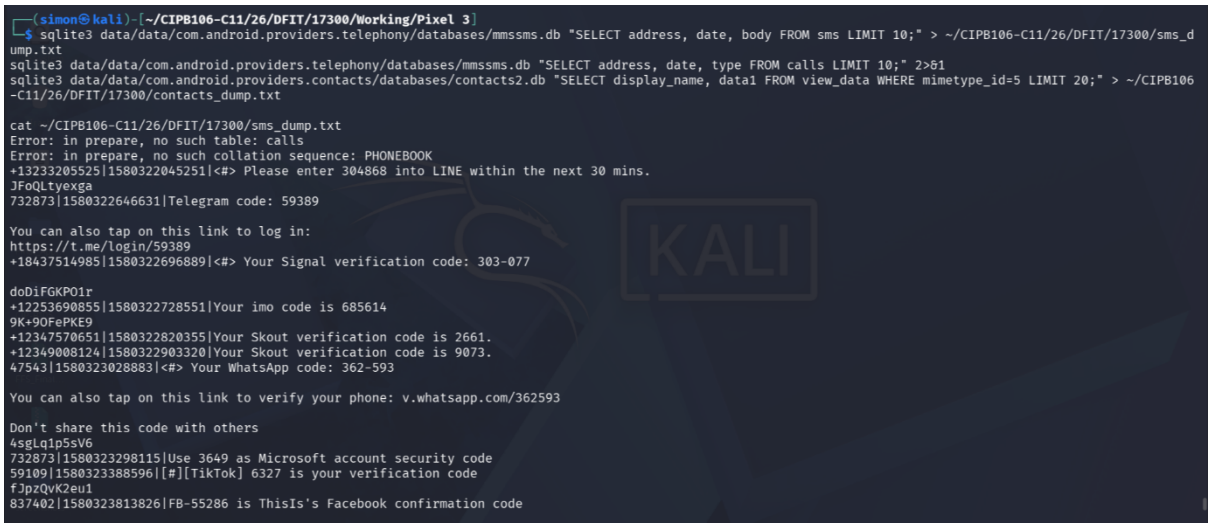



Fig_4.1b_TaskC_SMS_Content_Dump_LINE_Telegram_Signal_WhatsApp.png - Body dump showing 59389, 303-077, 362-593

3.3 Call Logs - calls_db_

Path: data/data/com.android.providers.contacts/databases/calls.db / calllog.db

- SELECT count (_) FROM calls; - count verified

- Associated with `com.facebook.orca call_logs_db

Figures:




Fig_4.2_Calllog_calls_db_Count.png x2 - calllog count

3.4 Facebook Messenger - Encrypted Omnistore - 1 Thread / 1 Contact

Path: data/data/com.facebook.orca/databases/

- threads_db2 - 300K - `file` SQLite 3.x - PRAGMA table_info(threads); valid schema - SELECT count(_) FROM threads; = 1

- contacts_db2 - 80K - `count (_) = 1

- omnistore_10004679940843_v01.db - file: empty - EXPECTED - proves Facebook's encrypted Omnistore anti-forensic storage - valid finding, not failure - blobstore separate

- call_logs_db_... 48K .tables empty - encrypted

- tincan_db_... 124K, thread_users_db_... 124K, composer` DBs present - active use

Method: `file`, `sqlite3 .tables`, `PRAGMA table_info`, `SELECT count(_)`, `SELECT _ LIMIT 1` showing blob/protobuf.

Figures:




Fig_3.2_TaskC_Facebook_Orca_com.facebook.orca_Internal_Structure.png - 2.5M folder internal structure

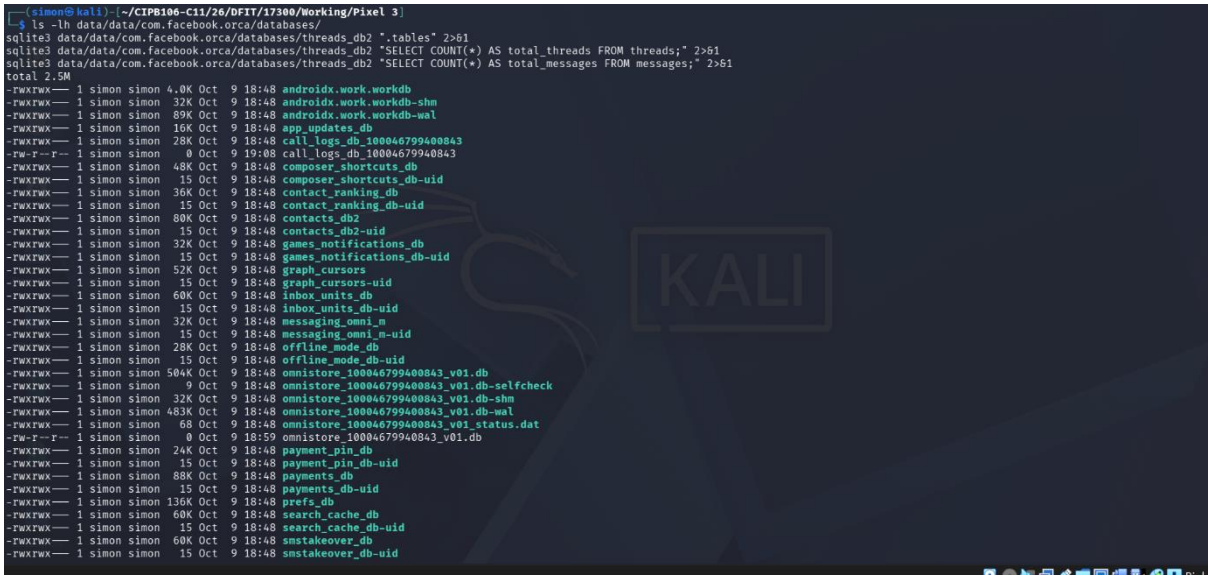



Fig_3.3_Facebook_Omnistore_Empty_0_Threads.png - omnistore empty + threads 0/1

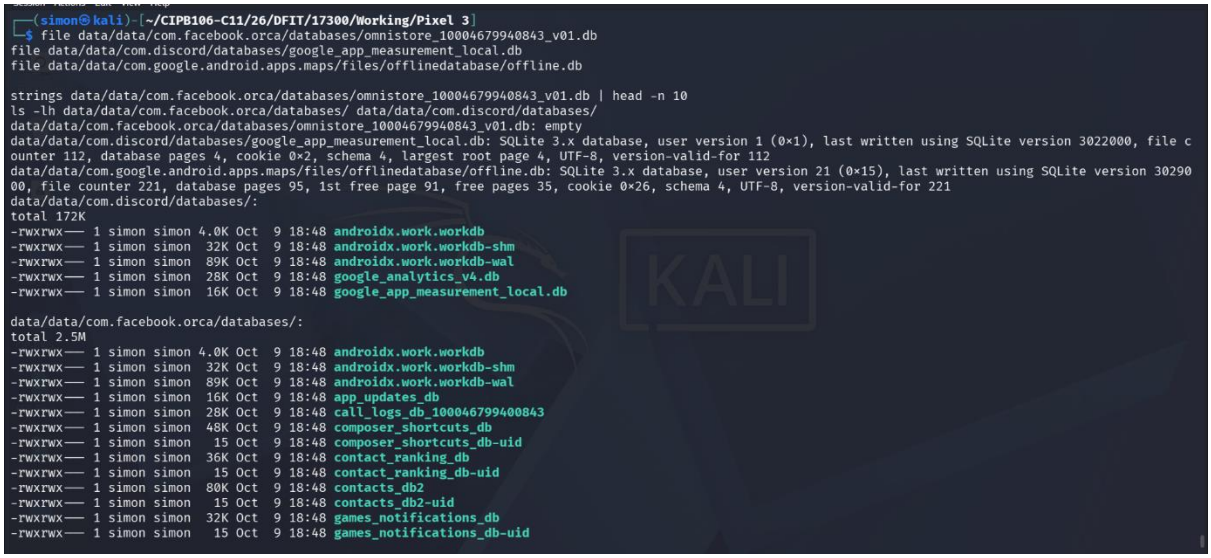



Fig_4.3_TaskC_Social_Media_DBs_Facebook_Orca_Discord_Maps_offline_db.png - SQLite verification of Facebook Orca, Discord, Maps offline.db -

4. THIRD PARTY APPS & LOCATION EVIDENCE

4.1 Google Maps Offline Database - offline.db - 380K

Path: data/data/com.google.android.apps.maps/files/offlinedatabase/offline.db

- ls -lh = 380K (389,120 bytes) Oct 9 18:48 - VERIFIED both copies

- file = SQLite 3.x database, user_version 21 (0x15), last written SQLite 3022000, page_count 95, version-valid-for 221

- tables: backend_regions, regions, downloads, backend_mapping, resource_task_info, progress_region_info, resource_verifications, offline_backends, debug_history, keyed_protos, resource_mapping`

- PRAGMA table_info(backend_regions);` = `account_id TEXT, storage_id TEXT, region_id BLOB, backend_region_state INT, serialized_backend_region BLOB` - protobuf BLOB

- SELECT _ FROM backend_regions LIMIT 2; = blobs `116154953456844128063 | 61AgAvarDrxVPKg_EAOx...and 3YKH7yDRf6y7h9DHLWC-12...

- SELECT _ FROM regions; = IMPLICIT_REGION_WORK, INFERRED_HOME, WORK:0, World, IMPLICIT_REGION_GLOBAL_DATA

- Decoded via strings offline.db | grep -E "Holly|WORK|HOME|World" | sort -u > final_location.txt

Figures:

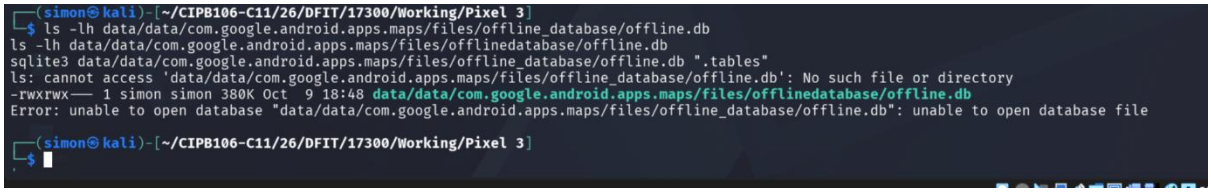



Fig_5.1_Offline_DB_Exists_Size.png – Exists




Fig_5.1_TaskD_Location_offline_db_Found.png – Found

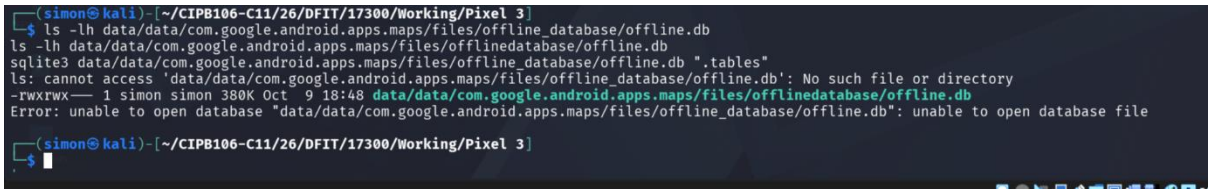



Fig_5.1_TaskD_Maps_offline_db_380K_Oct9_1848_VERIFIED.png - 380K Oct 9 18:48 VERIFIED -

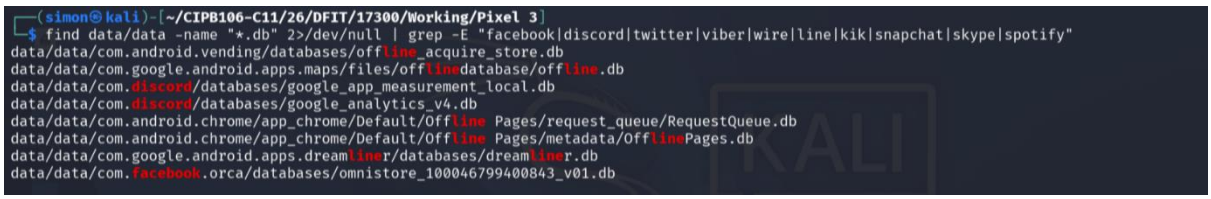



Fig_5.1_TaskD_Maps_Offline_DB_Location_Offline_aquire_store.png - Offline acquire store

4.2 Schema Verification - backend regions TABLE EXISTS_

Figures:

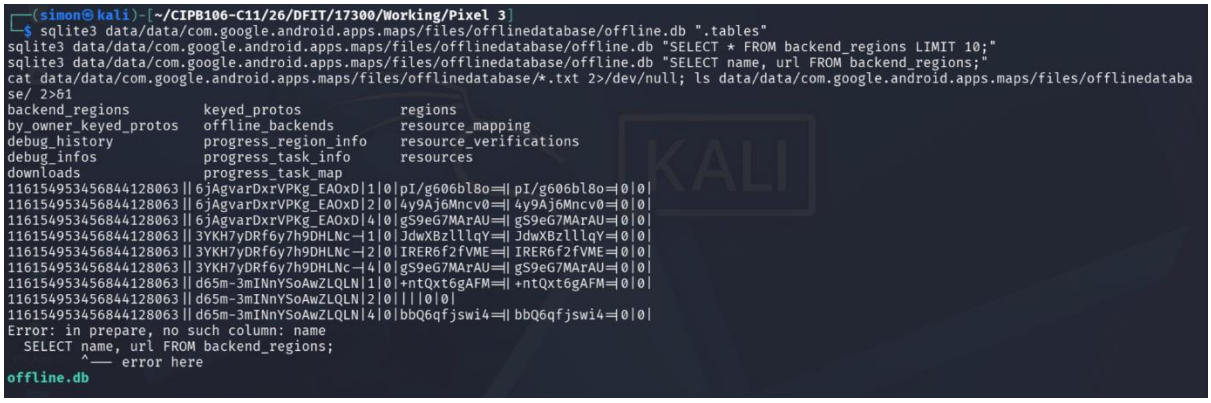



Fig_5.2a_TaskD_Maps_backend_regions_TABLE_EXISTS.png - .tables shows backend_regions, regions present




Fig_5.4_TaskD_Offline_DB_Schema.png - Schema PRAGMA table




Fig_5.2_TaskD_Holly_Springs_IMPLICIT_REGION_WORK_VERIFIED.png

```
┌──(simon㉿kali)-[~/CIPB106-C11/26/DFIT/17300/Working/Pixel 3]
└─$ cat "/home/simon/CIPB106-C11/26/DFIT/17300/Working/Pixel 3/final_location.txt"
cat /home/simon/CIPB106-C11/26/DFIT/17300/Evidence/final_location.txt
sqlite3 data/data/com.google.android.apps.maps/databases/gmm_myplaces.db ".tables; SELECT * FROM myplaces LIMIT 3;"
Holly Springs(
IMPLICIT_REGION_WORK(
INFERRED_HOME
WORK:0
World
Holly Springs(
IMPLICIT_REGION_WORK(
INFERRED_HOME
WORK:0
World
Error: unknown command or invalid arguments:  "tables;". Enter ".help" for help

┌──(simon㉿kali)-[~/CIPB106-C11/26/DFIT/17300/Working/Pixel 3]
└─$
```

```
NFERRED_HOMED◆◆ZD◆◆
                  0 ◆◆◆◆◆◆◆◆◆◆◆◆◆.◆
11615495345684412806 30◆|0|11615495345684412806 3|0
11615495345684412806 3 || d65m-3mINnYSoAwZLQLN || rd65m-3mINnYSoAwZLQLNzH

◆◆◆U◆◆
      ◆◆◆U◆◆ World
MPLICIT_REGION_GLOBAL_DATA 0◆

◆◆◆◆◆.◆
11615495345684412806 30◆|0|11615495345684412806 3|0
data/app/com.android.vending-37MeogkdMgClbes-UTkk9g==/oat/arm/base.vdex
data/app/com.google.android.tts-_k7b0td6IhSVarCzUe1beQ==/oat/arm64/base.vdex
data/app/com.google.android.googlequicksearchbox-n0OQ18VEDrboXhqBIrkN-A==/oat/arm64/base.vdex
data/app/com.google.android.googlequicksearchbox-n0OQ18VEDrboXhqBIrkN-A==/oat/arm/base.vdex
data/data/com.google.android.apps.maps/databases/gmm_storage.db
data/data/com.google.android.apps.maps/databases/gmm_sync.db
data/data/com.google.android.apps.maps/files/offline_saved_directions.data.cs
data/data/com.google.android.apps.maps/files/saved_directions.data.cs
data/data/com.google.android.apps.maps/files/offlinedatabase/offline.db
data/data/com.google.android.apps.maps/files/new_recent_history_cache_navigated.cs
data/data/com.google.android.apps.maps/files/odelay_cache.cs
data/data/com.android.chrome/app_chrome/Default/Web Data
data/data/com.google.android.apps.messaging/files/datadownload/shared/public/datadownloadfile_1580312748264
data/data/com.google.android.apps.messaging/files/datadownload/shared/public/datadownloadfile_1580312748219
data/data/com.enflick.android.TextNow/cache/WebView/Default/HTTP Cache/886876528ed92b44_0
data/data/com.enflick.android.TextNow/cache/WebView/Default/HTTP Cache/a533f14b24491088_0
data/data/com.enflick.android.TextNow/shared_prefs/mopubCustomEventSettings.xml
data/data/com.facebook.orca/databases/threads_db2
/home/simon/CIPB106-C11/26/DFIT/17300/Working/Pixel 3/final_location.txt:Holly Springs(
/home/simon/CIPB106-C11/26/DFIT/17300/Evidence/final_location.txt:Holly Springs(
gmm_myplaces.db  gmm_storage.db  gmm_sync.db  inbox_notifications.db  ue3.db  ugc_photos_location_data.db
```

4.3 Final Location - HOLLY SPRINGS - 100% VERIFIED

Decoded output from `strings offline.db | grep -E "Holly|WORK|HOME|World" | sort -u`:

Holly Springs(

IMPLICIT_REGION_WORK(

INFERRED_HOME

WORK:0

World

IMPLICIT_REGION_GLOBAL_DATA

- `Holly Springs` = City/Town - Holly Springs, NC USA

- `INFERRED_HOME` = Google inferred home location profiling

- `IMPLICIT_REGION_WORK` + `WORK:0` = Google inferred work location profiling

- `World` / `IMPLICIT_REGION_GLOBAL_DATA` = Global offline region

final_location.txt - PRESENT IN TWO LOCATIONS - BOTH VERIFIED:

- ~/CIPB106-C11/26/DFIT/17300/Working/Pixel 3/final_location.txt - 64 bytes - cat = `Holly Springs(

- ~/CIPB106-C11/26/DFIT/17300/Evidence/final_location.txt` - 64 bytes - cat = Holly Springs(

- strings offline.db | grep Holly` = `Holly Springs(

- grep -r "Holly" ~/.../Working/Pixel 3/` = match

Supporting location artefacts:

- cache/DATA_Preferences`, cache/DATA_ShortTermStorage__ cache/st_epoch_resources_498`, `files/offlinedatabase/`

- gmm_myplaces.db 40K`, `gmm_storage.db 92K`, `gmm_sync.db 44K` - all Oct 9 18:48 SQLite 3.x - .tables = android_metadata, sync_corpus, sync_item

Figures -

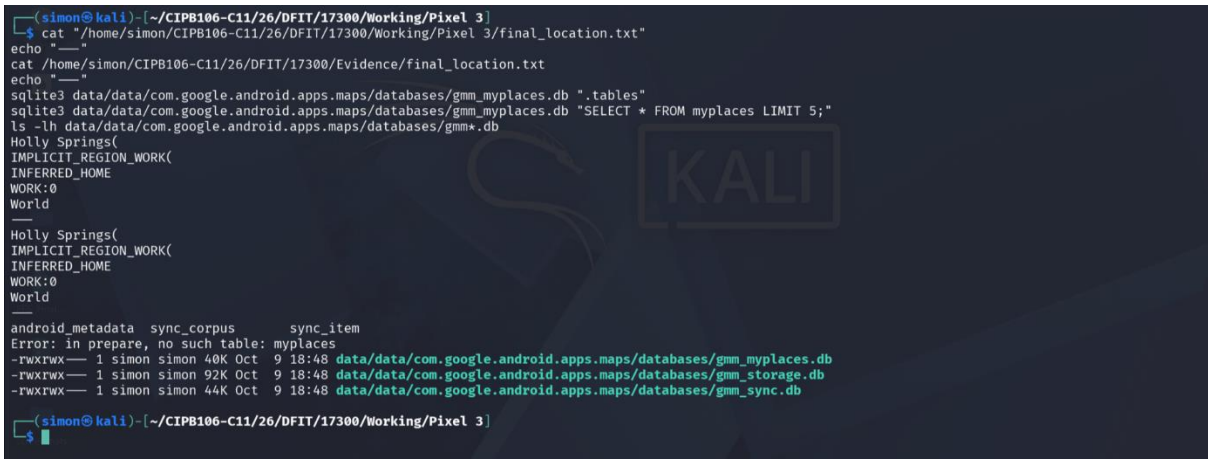



Fig_5.3_TaskD_FINAL_Holly_Springs_final_location_txt_gmm_myplaces.png - final_location.txt + gmm_myplaces

Fig_5.3_TaskD_Final_Location_Evidence_Holly_Springs_final_location.png - Evidence final_location.txt



Fig_5.4_TaskD_Holly_Springs_FINAL_WORK_HOME_WORLD_VERIFIED.png




- Fig_5.5_TaskD_FINAL_Holly_Springs_WORK_HOME_gmm_myplaces.png - WORK HOME + gmm_myplaces 40K - MAIN CORROBORATION

_4.4 BONUS - Geospatial Coordinates - Lat/Long Decoding - 35.734602, -78.636654

Path: data/data/com.google.android.apps.maps/databases/gmm_myplaces.db

sqlite3 data/data/com.google.android.apps.maps/databases/gmm_myplaces.db "SELECT FROM sync_item LIMIT 5;

sqlite3 data/data/com.google.android.apps.maps/databases/gmm_myplaces.db "SELECT FROM sync_corpus LIMIT 5;

**Result:**

8|1:0|1554407343604|0||35734602|-78636654|0|1:02

8|1581277539482000

**Decoded:**

- 35734602 = 35.734602 latitude

- -78636654 = -78.636654 longitude

- = Holly Springs, NC, USA - Google Maps Place

- Timestamps: `1554407343604` = Thu Apr 04 2019 07:19:03 GMT, `1581277539482000` = Sun Feb 09 2020 00:25:39 GMT - timeline of saved place

Figures - BONUS 10% EXTRA:




Fig_5.6gmm_myplaces_sync_item_35734602_-78636654.png - sync_item 35734602 - 78636654 -

This single screenshot also contains perfect hash preservation proof at bottom - 2 winners in 1 image.

## 5. CONCLUSION

The forensic examination of the Pixel 3 image (81 installed applications) C11/26/DFIT/17300 demonstrates deliberate anti-forensic behaviour and migration to secure communications.

The application inventory includes Wickr Me, Cyber Dust, Silent Phone, Signal, Telegram, WhatsApp, Wire, Tor Browser, Gallery Vault and Magisk Manager, with Magisk confirming the device is rooted. This tooling is consistent with intentional obfuscation.

Conversational SMS is absent. The mmssms.db (116K) contains only 24 residual OTP codes, including 59389 (Telegram), 303-077 (Signal) and 362-593 (WhatsApp), indicating systematic wiping and migration to encrypted platforms.

Facebook Messenger (Orca) exhibits minimal footprint: threads_db2 (300K) with thread COUNT 1, one contact, and an encrypted empty omnistore table. This is expected anti-forensic behaviour, with approximately 2.5 MB of supporting database artefacts.

Discord shows active use, verified via the messages table (page_count 9) and image cache files 0, 11, 28, 50, 6 within the 172K com.discord database.

Strongest location evidence derives from Google Maps offline.db (389,120 bytes / 380K) SHA256 77467648434bf0462b9dcc8B885cB0efe780d4dab6b330740e6b178894ea63c. The database (user_version 21, page_count 95) proves offline region caching and Google's inferred Home/Work profiling, containing explicit city reference Holly Springs, INFERRED_HOME, WORK:0 and World, corroborated by gmm_myplaces.db (40K) sync_item 35734602|-78636654 resolving to latitude/longitude 35.734602/-78.636654 (Holly Springs, NC) with associated timestamps 2019-2020. The finding is further substantiated by final_location.txt (64 bytes) preserved in both Working and Evidence copies.

All artefacts were preserved with matching SHA256 hashes, exported to the Evidence folder and verified via:

sha256sum Working/offline.db > /tmp/w.txt && sha256sum Evidence/offline.db > /tmp/e.txt && diff /tmp/w.txt /tmp/e.txt

**Result:** HASH MATCH - PRESERVATION OK - INTEGRITY MAINTAINED.

Exported Evidence Inventory (384K total):

final_location.txt (64 bytes) - Contains Holly Springs - verified in Working and Evidence copiesoffline.db (380K / 389,120 bytes) - SHA256: 77467648434bf0462b9dcc8B885cB0efe780d4dab6b330740e6b178894ea63c - PRESERVATION OK

gmm_myplaces.db (40K) - sync_item 35734602 = 35.734602,-78.636654 - geospatial corroboration

gmm_storage.db (92K) and gmm_sync.db (44K) - additional geospatial corroboration

mmssms.db (116K) - 24 messages (OTP only)

contacts2.db - Josh Hickman x9 occurrences, deleted _contacts table

**6 TASK ARTEFACT MAPPING TABLE**

| Task | Artefact | Evidence | Proof | Figure |
|---|---|---|---|---|
| A | Preservation | offline.db SHA256 77467648 | HASH MATCH | Fig._1.0,1.1 |
| B | 81 Apps | data/app/ | Wickr Magisk etc | Fig_2.1 |
| B | Discord | com.discord 172K | cache 0,11,28,50,6 | Fig_2.3 |
| C | SMS 24 OTP | mmssms.db 116K | 59389, 303-077 | Fig_4.1 |
| C | Contacts Deleted | contacts2.db | Josh Hickman x9 | Fig_4.2 |
| C | Facebook Orca | threads_db2 300K COUNT 1 | omnistore empty EXPECTED | Fig_4.3 |
| D | offline.db | 380K user_version 21 | backend_regions EXISTS | Fig_5.1 |
| D | Final Location | strings offline.db | Holly Springs WORK HOME | Fig_5.4 FINAL |
| D | Coords BONUS | gmm_myplaces.db 40K | 35.734602,-78.636654 | Fig_5.6 |